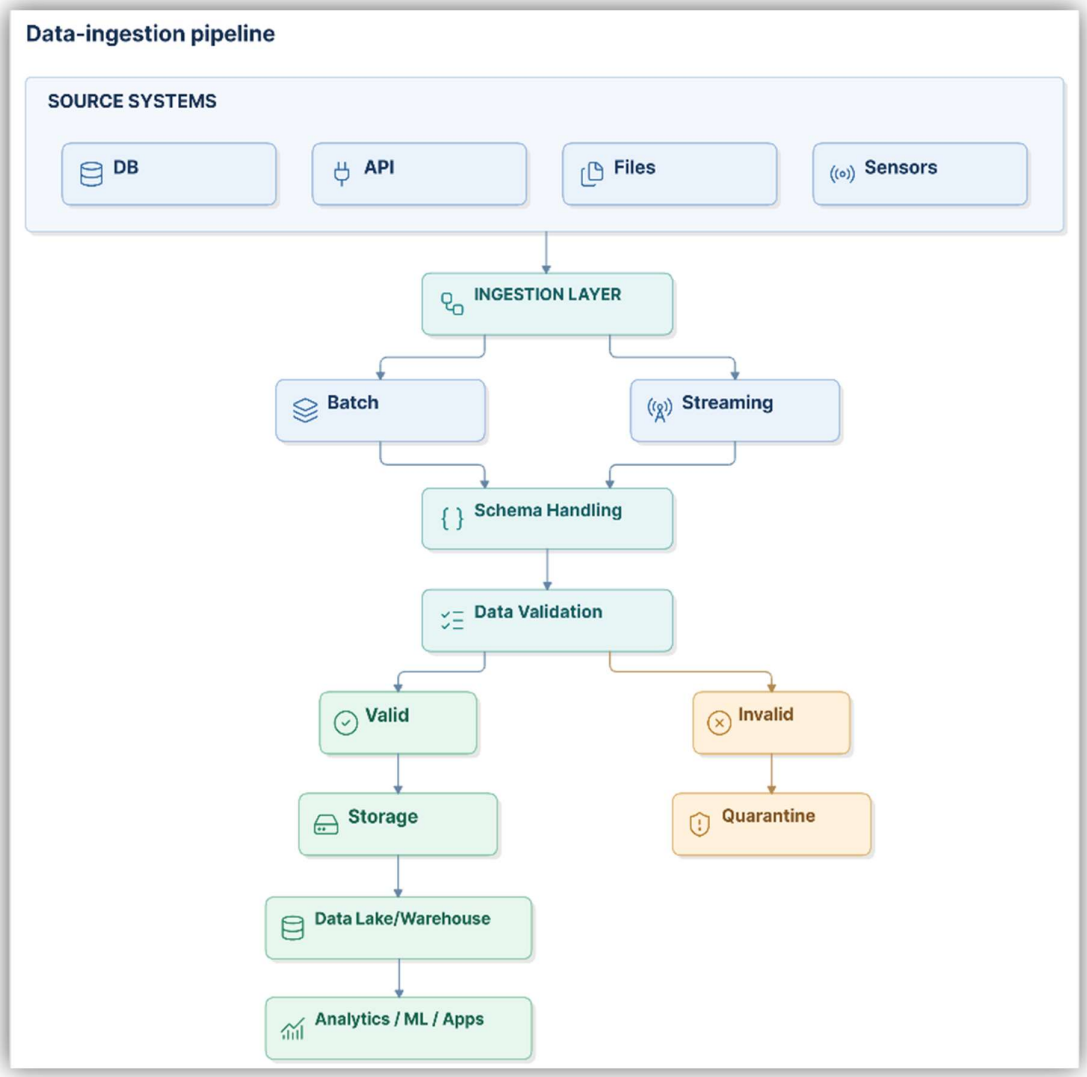
INGESTION

Things to ponder:

- **Source system** = where data originates.
- **Schema** = structure/blueprint of data.
- **Batch** = data moved in groups.
- **Streaming** = data moved continuously/as events occur.
- **Full load** = process all data.
- **Incremental load** = process new/changed data.
- **Validation** = check whether data meets expected rules.
- **Latency** = time taken for data to become available.
- **Freshness** = how current the available data is.
- **Error handling** = detect, retry, log, quarantine, alert, and recover from failures.



Arising Questions:

Topic	Core question
Source system analysis	Where is the data coming from and what is it like?
Batch vs streaming	How quickly does data need to arrive?
Schema handling	What structure does the incoming data have?
Data validation	Is the incoming data acceptable?
Incremental loading	Can we load only new/changed records?
Latency/freshness	How current does the data need to be?
Error handling	What happens when ingestion fails?

Main types of validation:

1. **Completeness Check**: Is required data present?
2. **Validity checks**: Is the value allowed?
3. **Format checks**: Is the value in the expected format?
4. **Range checks**: Is the value within an acceptable range?
5. **Uniqueness checks:** Is a field supposed to be unique?
6. **Referential integrity checks**: Does this record refer to something that actually exists?

Parameters	***Full Load***	***Incremental***
Reads	Everything	New/changed data
Simplicity	Simple	More complex
Cost	Higher for large data	Lower
Speed	Slower as data grows	Usually faster
Good for	Small datasets / initial load	Large changing datasets
Risk	Reprocessing everything	Missing changes if tracking is wrong

Project

QuickCare – The Data Challenge

1. What problems does QuickCare have?

Scenario problem	What it means	Required solution
Clinic information shared at end of day	Not real-time	**Batch ingestion**
Book/cancel/delay events needed as they happen	Low-latency operational data	**Streaming ingestion**
App data changed after a recent app update	Source schema may have evolved	**Schema-change detection + mapping**
Some records are incomplete	Missing/invalid data	**Validation + quarantine/rejection**
Appointments appear more than once	Duplicate ingestion/events	**Deduplication + idempotent processing**
Clinics correct information already shared	Existing records can change	**Incremental loading + upsert/update handling**
Different teams need different freshness	Not every workload needs real-time data	**Different ingestion/freshness paths**
Avoid repeatedly processing unchanged data	Full reload is wasteful	**Incremental ingestion**
Clinic information arrives late	Late-arriving data	**Monitoring + watermark/catch-up/reprocessing**

1. **Recommended ingestion approach**

Ans: Clinic systems → Batch
Clinic information is normally shared **at the end of the day.**

2. **Appointment booking system**

Ans: Streaming
Bookings, cancellations, and delays need to be available as they happen.

3. **Schema problem: how to rectify it.**

Ans: Do not let every source's changing structure directly dictate the central platform.

QuickCare – Ingestion Recommendation

1. Source Systems

- Clinic systems (clinic info, daily batch)
- Appointment booking system (booked/cancelled/delayed – real-time)
- Patient mobile app (contact & appointment updates – real-time)

Key considerations: understand data types, update frequency, data quality issues, and any format or schema differences (e.g., new app fields).

2. Batch vs. Streaming

- Clinics → Batch (end of day)
- Appointments → Streaming (near real-time)
- Mobile app → Streaming (near real-time)

Use streaming for operational needs (bookings, cancellations, delays, app updates). Use batch for daily clinic data.

3. Schema Handling

- Maintain a canonical schema in the data platform.
- Map source schemas to the canonical model (handling new app fields).
- Detect schema changes and use schema evolution (additive changes OK; breaking changes require review).

4. Data Validation

- Check for missing values (e.g., patient ID, appointment time).
- Validate formats (e.g., phone, date, email).
- Check for consistency (e.g., appointment status, time ranges).
- Flag unexpected values (e.g., new app fields, invalid codes).

5. Incremental Loading

- Use timestamps or change data capture (CDC) to load only new/changed records.
- Detect updates (e.g., clinic corrections) and apply upserts (update/insert).

Full load only for initial setup or major schema changes.

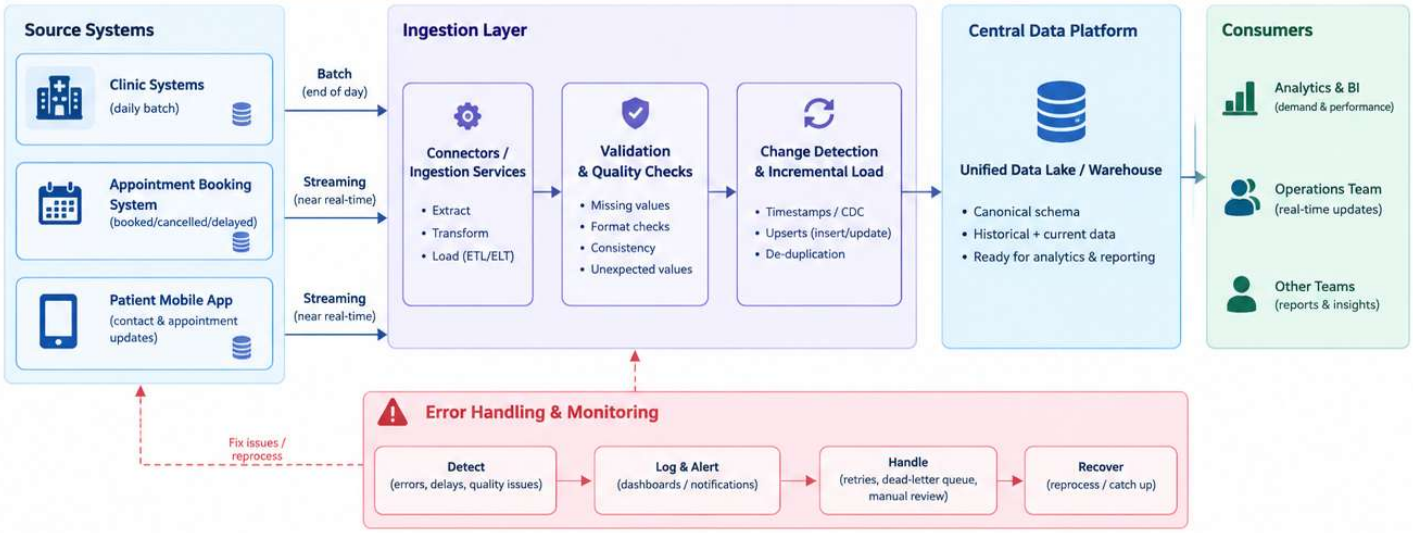
6. Latency & Freshness Trade-offs

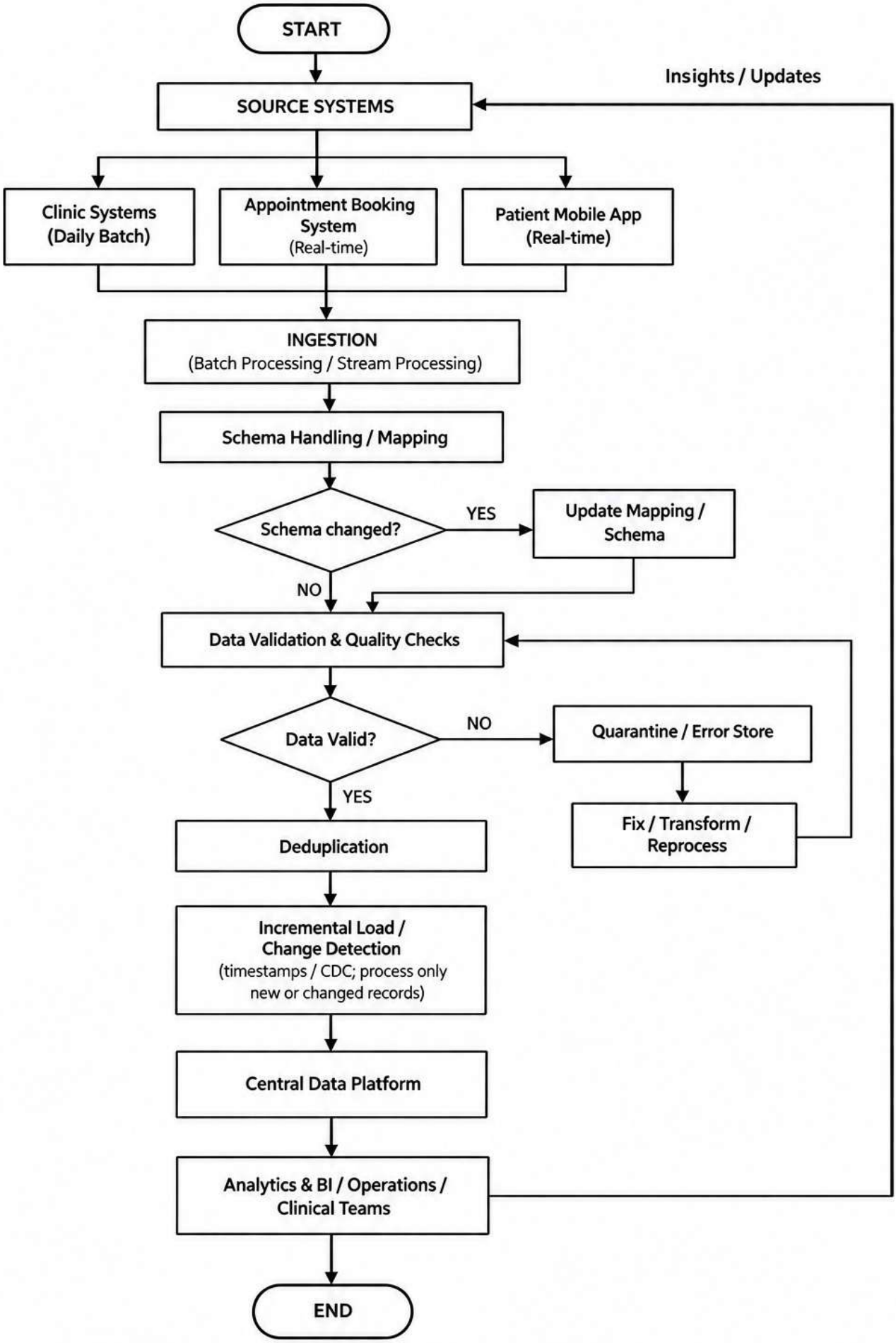
- Real-time needs (operations) → low latency, higher complexity (streaming).
- Daily reporting (demand, performance) → batch, lower complexity.
- Balance freshness with cost, processing load, and system complexity.

7. Error Handling

- Common failures: connection issues, schema mismatches, validation errors, duplicates, late/ missing data.
- Detect: monitoring, data quality checks, error logs.
- Handle: retries, dead-letter queue, alerting, and manual review for complex cases.
- Recover: reprocess failed records / run incremental catch-up.

Proposed Ingestion Flow




START
Insights / Updates
SOURCE SYSTEMS
Clinic Systems (Daily Batch)
Appointment Booking System (Real-time)
Patient Mobile App (Real-time)
INGESTION (Batch Processing / Stream Processing)
Schema Handling / Mapping
Schema changed?
YES
Update Mapping / Schema
NO
Data Validation & Quality Checks
Data Valid?
NO
Quarantine / Error Store
Fix / Transform / Reprocess
YES
Deduplication
Incremental Load / Change Detection (timestamps / CDC; process only new or changed records)
Central Data Platform
Analytics & BI / Operations / Clinical Teams
END